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Effect of electrochemical topology on detection sensitivity in MEA assay for drug-induced cardiotoxicity screening

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ABSTRACT

Cardiotoxicity remains a major challenge in drug development, accounting for 45% of medication withdrawals due to cardiac ischemia and arrhythmogenicity. To overcome the limitations of traditional multielectrode array (MEA)-based cardiotoxicity assays, we developed a Nafion-coated NanoMEA platform with decoupled reference electrodes, offering enhanced sensitivity for electrophysiological measurements. The 'Decoupled' configuration significantly reduced polarization resistance (Rp) from 12.77 MΩ to 3.41 MΩ, improving charge transfer efficiency as demonstrated by electrochemical impedance spectroscopy and cyclic voltammetry. Additionally, the limit of detection significantly decreased from 0.175 MΩ (Coupled) to 0.040 MΩ (Decoupled), underscoring the system's enhanced sensitivity. Using human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs), we evaluated the effects of three proarrhythmic drugs: Ranolazine, Domperidone, and Sotalol. Under the decoupled condition, the platform exhibited reductions in IC50 values for Domperidone (0.71 μM–0.29 μM), Sotalol (7.61 μM–0.27 μM), and Ranolazine (53.08 μM–5.89 μM), demonstrating significantly improved drug detection sensitivity. Longitudinal analysis revealed significant alterations in key electrophysiological parameters, including beating period (BP), field potential duration (FPD), spike slope, and amplitude, which were consistent with the known pharmacological actions of these drugs. Further validation through action potential (AP) waveform analysis showed enhanced repolarization dynamics, confirming the platform's predictive capabilities. Our findings highlight the critical role of electrochemical topology in optimizing MEA performance. The NanoMEA system, featuring decoupled Nafion-coated electrodes, represents a robust and sensitive platform for cardiotoxicity screening, setting a new standard for preclinical drug safety assessment and advancing bioelectronic device design for cardiac research.

1. Introduction

Cardiotoxicity remains a significant challenge in safety pharmacology during the drug development process. Between 1994 and 2006, cardiotoxicity was responsible for 45% of all drug withdrawals, primarily due to adverse effects of cardiac ischemia and arrhythmogenicity (Jones and Ewer, 2006). Current screening methods for cardiotoxicity rely predominantly on in vitro and in vivo models. However, these approaches are constrained by challenges such as interspecies variation (Perez et al., 2019) and limited translational relevance

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(Saleem et al., 2020). These shortcomings often lead to false positives or negatives, reducing the reliability of safety assessments and posing significant hurdles to the development of safer drugs.

The advent of human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) has revolutionized cardiotoxicity screening (Schwach et al., 2020). In vitro models, particularly those employing hiPSC-CMs, are increasingly recognized for their ability to provide more biologically relevant insights compared to conventional animal models (Macadangdang et al., 2015; Choi et al., 2020; Huang et al., 2021). However, despite their significant advantages, traditional detection methods, such as single-electrode systems, often fail to capture the complex and dynamic electrophysiological responses of hiPSC-CMs (Cardinale et al., 2020; Stella Stoter et al., 2020).

Microelectrode array (MEA) assays employing hiPSC-CMs have emerged as a powerful tool for high-resolution, non-invasive monitoring of electrical activity in cultured cardiomyocytes (Navarrete et al., 2013; Blinova et al., 2018). MEAs enable detailed characterization of drug-induced electrophysiological alterations, providing valuable insights into ion channel function, conduction properties, and cellular metabolism (Choi et al., 2021a). However, traditional MEA platforms face critical limitations, such as signal degradation caused by electrode heat, susceptibility to ambient noise, and the rigidity of fixed electrodes, which may influence the physiological behavior of cardiomyocytes.

To address these limitations, we recently introduced a novel nanopatterned Nafion-coated microelectrode array device, termed NanoMEA (Choi et al., 2020, 2021a,b; Smith et al., 2020). Nafion, a proton-conductive, ion-permeable polymer, offers unique advantages, including its ability to support nanoscale 3D patterning and optical transparency for post hoc cellular analysis (Choi et al., 2022, 2024). The NanoMEA system integrates these features with Local Extracellular Action Potential (LEAP) analysis (Hayes et al., 2019; Melle et al., 2020), designed to enhance the alignment and maturation of hiPSC-CMs, thereby improving the reliability and sensitivity of electrophysiological assessments. Despite these advancements, challenges remain, particularly the double-layer effect, a phenomenon occurring at the electrode-electrolyte interface. This effect results from the accumulation of opposite charges across the interface, leading to excessive capacitance. While double-layer capacitance enhances charge storage, it can also introduce phase delays, distort signal linearity, and mask subtle electrophysiological responses. These issues are particularly pronounced when high ionic conductivity materials, such as Nafion, are used, as they facilitate rapid ion transport but may exacerbate capacitance-related artifacts. Addressing these limitations are critical to ensuring accurate and reliable electrophysiological measurements.

To address this challenge, we investigated how electrochemical topology affects the sensitivity and reliability of electrophysiological measurements. Specifically, we proposed a novel modification involving the decoupling of the reference electrode during the application of Nafion resin. This approach was designed to mitigate the adverse effects of double-layer capacitance, such as phase delays and signal distortions, by isolating the working electrode array from the reference electrode. By applying the Nafion resin exclusively to the working electrodes, we aimed to optimize charge transfer dynamics, reduce impedance, and enhance the linearity and sensitivity of the recorded signals.

This decoupled configuration not only mitigates the excessive capacitance but also creates a more uniform and stable electrode-electrolyte interface. Such improvements enable the NanoMEA system to detect subtle electrophysiological changes with greater precision, particularly in hiPSC-CMs exposed to proarrhythmic compounds. By refining the electrochemical topology, this study provides a blueprint for designing advanced bioelectronic platforms capable of overcoming the limitations of traditional MEAs while setting a new standard for sensitivity and reliability in cardiotoxicity assessments. To validate our improved assay and hypothesis, we focused on three proarrhythmic risk compounds: Ranolazine, Domperidone, and Sotalol. These drugs were meticulously selected to represent a spectrum of arrhythmia risks – high, medium, and low, respectively, as previously demonstrated (Blinova et al., 2018; da Rocha et al., 2020). Each compound was thoroughly analyzed at various dose levels in comparison to in-plate DMSO controls. This approach allowed for a comprehensive evaluation of key electrophysiological parameters, including beating period (BP), field potential duration (FPD), spike slope, and spike amplitude. By employing double subtraction methods for analysis, we could accurately discern drug-induced changes in these parameters. The study evaluated the drug toxicity of hiPSC-CMs using the NanoMEA system under both coupled and decoupled conditions. The coupled condition involves both the reference and working electrodes covered with Nafion resin, while the decoupled condition applies the Nafion resin exclusively to the working electrode array. Our results demonstrated significant alterations in electrophysiological parameters that correlated with the known pharmacological actions of these compounds on cardiomyocytes. These findings validate the enhanced NanoMEA assay with decoupling features and underscore its potential as a reliable and sensitive platform for detecting subtle effects of test compounds on hiPSC-CMs. By comparing these two configurations, the study highlights the superior sensitivity and reliability of the decoupled NanoMEA system, providing a significant improvement in the accuracy of cardiotoxicity assessments.

2. Experimental methods

2.1. Fabrication of Nafion patterns

Nanotopographical pattern features within the MEA platform were fabricated using capillary force lithography. First, nanogrooved polyurethane acrylate master molds were created using a replica molding process, as described previously (Choi et al., 2022, 2024; Smith et al., 2020). Polydimethylsiloxane (PDMS; Sylgard 184, Dow Corning) was then mixed at a 10:1 ratio (base: curing agent) and applied to the PUA mold, followed by curing overnight at 65 °C. Once cured, PDMS nanotopographic patterns were removed from the PUA mold, cut to the desired size, and pressed into a 2 μL drop of Nafion resin deposited onto the MEA substrate. Capillary action forced the Nafion solution to conform to the PDMS topography. The PDMS patterns were pressed onto Nafion-coated MEA surfaces and left to cure for 48 h at room temperature via solvent evaporation. All nanotopographic patterns were oriented vertically to ensure effective comparisons between wells. After curing, the PDMS molds were carefully removed, revealing thin films supporting the desired nanotopography.

2.2. Electrode performance evaluation using electrochemical impedance spectroscopy

The impedance properties of nanopatterned Nafion layers were measured using an Interface 1010E (Gamry Instrument) equipped with an Electrochemical Impedance Spectroscopy (EIS) module. Measurements were conducted over a frequency range of 1 kHz–100 Hz with an AC amplitude of 1 mV. For cyclic voltammetry, the potential range extended from -3V to 3V, which is sufficiently broad to encompass a wide range of redox processes for many electrochemical systems. The voltage sweep rate was set to 50 mV/s. The MEA electrode array was fabricated via photolithography. A negative photoresist was spin-coated and patterned under a UV mask aligner to define the electrode patterns. Subsequently, a thermal deposition process layered 20 nm of Cr and 200 nm of Au in sequence. The negative photoresist was lifted off by gently agitating the sample in acetone. The final electrode width and the spacing between electrodes were both 120 μm. To confine the Nafion and electrolyte solution on the interdigitated electrode area, PDMS wells were fabricated. A PDMS sheet was prepared by mixing PDMS prepolymer and curing agent at 10:1 wt ratio, then cured overnight at 65 °C. Individual wells were created using a 6 mm hole puncher on the cured PDMS sheet and attached to each MEA electrodes. To pattern the Nafion layer onto the electrodes, a 5% Nafion solution was dropped inside the

PDMS wells and pressed with a pre-patterned PDMS mold. After drying for 3 days, the PDMS mold was removed from the surface of electrode, leaving a nanopatterned Nafion layer (as described in Fabrication of Nafion Patterns section). The PDMS wells were filled with DPBS to wet the patterned Nafion layer. The sensitivity of the system was evaluated by calculating the Limit of Detection based on the polarization resistance (R_p) values obtained from EIS measurements. The limit of detection was determined using the standard signal-to-noise ratio method, defined as:

$$Limit\ of\ Detection = \frac{3 \cdot \sigma}{m}$$

where σ represents the standard deviation of baseline noise, and m denotes the sensitivity, calculated as the difference in mean R_p values between the coupled and decoupled electrode configurations.

2.3. hiPSC-CMs cell culture and seeding on NanoMEA

Commercially sourced hPSC-derived cardiomyocytes (hPSC-CMs) from Celogics were stored, thawed, and maintained according to the manufacturer's protocol. Cultures were kept at a 37 °C, 5% CO_2 environment throughout the culture period. To establish optimal cell culture conditions on nanoMEA devices, high-density cell solutions were drop-seeded directly onto the desired recording electrodes. A fibronectin coating (5 μg/mL) was applied to all MEA substrates as a 6 μL droplet, ensuring coverage of the recording electrodes. This coating was incubated for 1 h to enhance cell attachment. Distilled H_2O was placed around each MEA well to humidify the plate and prevent biopolymer evaporation. After the incubation period, the fibronectin solution was aspirated, and 17,500 cardiomyocytes, resuspended in 6 μL of medium, were carefully transferred to the biopolymer-coated electrodes. The MEAs were then incubated at 37 °C, 5% CO_2 for 3 h to facilitate cell attachment. Following this, additional medium was added to the MEA wells to support the cultured cells. Cultures were maintained for 21 days prior to analysis (unless otherwise stated), and medium was replaced every 2–3 days during this period.

2.4. Electrophysiological assessment of cardiomyocytes

Cardiac electrophysiological properties were assessed using 24-well MEA plates, each containing 16 electrodes per well (30 μm electrode diameter, 200 μm pitch), prior to (baseline) and 30 min after drug treatment. Measurements were performed using the Maestro MEA system (Axion Biosystems). During data acquisition, standard recording settings for spontaneous cardiac field potentials were applied using Axis software (version 2.1). Cells were maintained at 37 °C and 5% CO_2 throughout the 2-min recording period. The standard recording settings included a 130 × gain, a recording range of 1–25000 Hz, and a low-pass digital filter set of 2 kHz for noise reduction. For cardiomyocytes, the beat detection threshold was set at 300 μV, and the FPD detection threshold was 1.5 × noise. Field potential was analyzed with the platform software, generating the following outputs: beat period (ms), fast Na^+ slope (V/s), spike amplitude (mV), and FPD (ms). Reported results were calculated by averaging data from all electrodes in each well, followed by averaging data from duplicate wells. Data from drug-treated samples were normalized to baseline values and subsequently expressed as a percent change relative to vehicle-treated wells.

2.5. Immunocytochemistry of hiPSC-CMs on NanoMEA

To compare the effects of Nafion nano-patterned resin in decoupling and coupling conditions on the electrophysiological properties of hiPSC-CMs, cardiomyocytes were immunostained for markers of cardiac contractile machinery. Briefly, cells were fixed in 4% paraformaldehyde for 15 min and then blocked with 5% goat serum in PBS for 1 h at room temperature. Cells were then incubated with primary antibodies diluted in 1% goat serum in PBS overnight at 4 °C. The following day, cells were washed three times with PBS before being incubated in a secondary antibody solution. Counterstaining was performed using Vectashield containing DAPI (Vector Labs). Images were captured using the LEICA SP8 Confocal microscope platform with a 63x CFI Plan Apo NA1.2 objective lens. Images were acquired at a 12-bit resolution (1024x1024 pixels) with the pinhole set to 1 Airy unit for all images. The antibodies used in this study were as follows: mouse anti-α-sarcomeric actinin monoclonal antibody (1:200, Sigma-Aldrich), rabbit anti-connexin 43 monoclonal antibody (1:200, Sigma-Aldrich), Recombinant Alexafluor-647 conjugated cardiac Troponin T(cTnT) antibody (1:1000, abcam), Alexafluor-594 conjugated goat-anti-rabbit secondary antibody (1:500, Invitrogen), and Alexafluor-488 conjugated goat-anti-mouse secondary antibody (1:500, Invitrogen).

2.6. Drug treatment

All drug solutions, along with the DMSO control, were meticulously diluted using the same cell culture media used to maintain the cardiomyocyte cultures, ensuring consistency. The concentrations of drug solutions and the DMSO control, as outlined in Table S2, were prepared to span a range that encompasses clinically relevant doses and potential toxicity thresholds. To ensure maturation and stabilization of the cardiomyocyte cultures, baseline measurements of FPD, contractility, and action potential activity were performed using the Maestro MEA system at Days 7 and 14 post-cells seeding. Upon confirmation of culture stabilization, drug treatments were initiated at Day 21. Before administering drug solutions, a final set of baseline measurements for FP and beat period was conducted to establish a reference point for subsequent analyses. Each well then received 30 μL of the respective drug solutions according to a pre-defined plate map, ensuring uniformity across experiments. Following a 30-min incubation at 37 °C and 5% CO_2, measurements of field potential, contractility, and action potential activity were obtained. All recordings were carefully documented and analyzed to evaluate the effects of the drugs on the electrophysiological properties of the CM cultures.

3. Results & discussion

3.1. Design of the NanoMEA platform with decoupled electrochemical topology

The nanopatterned Nafion-coated microelectrode array device, called NanoMEA (Smith et al., 2020), comprises four equidistant working electrodes arranged in a square pattern, with reference electrodes positioned around these working electrodes (Fig. 1A). We categorized the electrode configurations as 'coupled' (both reference and working electrodes coated with Nafion) and 'decoupled' (Nafion coating applied exclusively to the working electrodes). The primary objective was to assess how these configurations influence the sensitivity and linearity of the NanoMEA in capturing electrophysiological signals (Fig. 1B). hiPSC-CMs were plated on NanoMEA devices under both configurations (Fig. 1C). By isolating the working electrode array from the reference electrode, the decoupled configuration mitigated double-layer capacitance effects, thereby reducing phase distortions and enhancing signal linearity. These advantages are clearly illustrated by the linear-phase signals recorded with the decoupled setup compared to the non-linear distortions observed in the coupled configuration. To further evaluate the impact of Nafion patterns and electrode configurations, we assessed the adhesion, topology, and protein expression of hiPSC-CMs on MEA plates. Immunocytochemistry images (Fig. 1D) revealed robust alignment of hiPSC-CMs on nanopatterned Nafion-coated surfaces, as evidenced by the organized expression of α-sarcomeric actinin (α-SA), cardiac Troponin-T (cTnT), and Connexin 43 (CX43). Cells cultured on patterned surfaces exhibited directional alignment consistent with the Nafion pattern, whereas those on

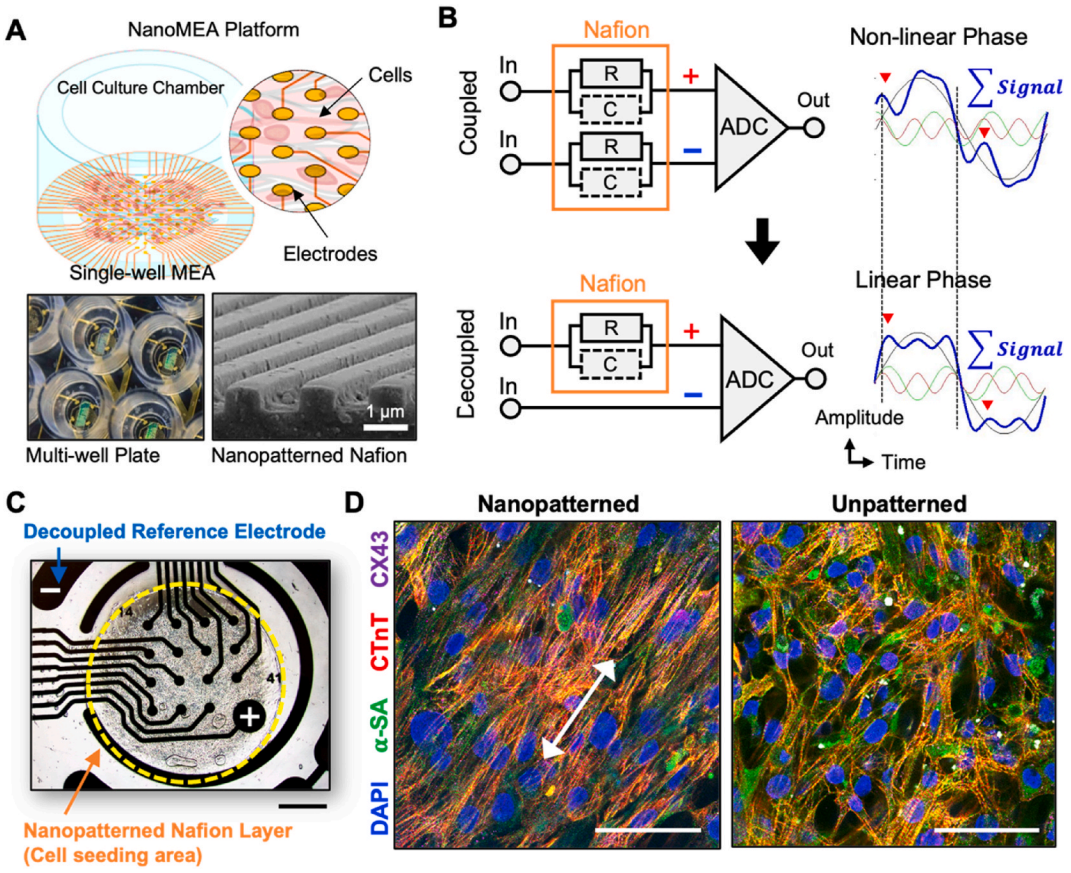


Fig. 1. The NanoMEA with decoupled reference electrodes for enhanced signal linearity. (A) Graphical overview and photograph of the NanoMEA and SEM image of Nafion nanotopography (Scale bar: 1 μm), (B) Equivalent circuit diagram illustrating the reference electrode decoupling scheme, designed to improve signal linearity at the cell-Nafion interface. (C) Photograph of the NanoMEA device featuring decoupled reference electrodes. (Scale bar: 500 μm) (D) Alignment of iPSC-derived cardiomyocytes cultured on the nanopatterned Nafion layer for 2 weeks (Scale bars: 50 μm). Double-headed arrow indicates the direction of cellular alignment.

unpatterned surfaces showed random orientations. Notably, the decoupled configuration, combined with the nanopatterned Nafion layer, further enhanced cell alignment and functional organization, underscoring the synergistic effect of the electrochemical and topographical optimizations.

3.2. *Effects of decoupled electrochemical topology on charge transfer efficiency*

EIS and cyclic voltammetry analyses provided a detailed assessment of electrode performance under both 'coupled' and 'decoupled' configurations (Figs. 2 and 3A–D, Table S1). These assessments offered a multifaceted perspective on the enhancement of charge transfer efficiency resulting from decoupling treatments. The physical interpretations of EIS data for electrodes, previously reported by Mei et al. (2018), were adopted to analyze the results. The EIS results (Fig. 3A) revealed a distinct reduction in impedance across the frequency range for 'Decoupled' electrodes compared to their 'Coupled' counterparts, indicating significantly improved charge transfer efficiency. This robust performance demonstrates the 'Decoupled' configuration's ability to

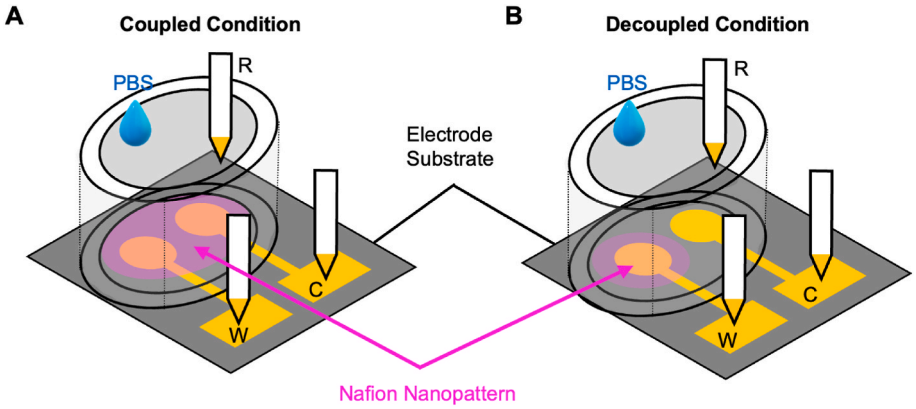


Fig. 2. Schematic representation of the electrochemical impedance spectroscopy (EIS) setup for measuring cyclic voltammetry and impedance under 'coupled' and 'decoupled' conditions. The setup includes the reference electrode (R), working electrode (W), and counter electrode (C).

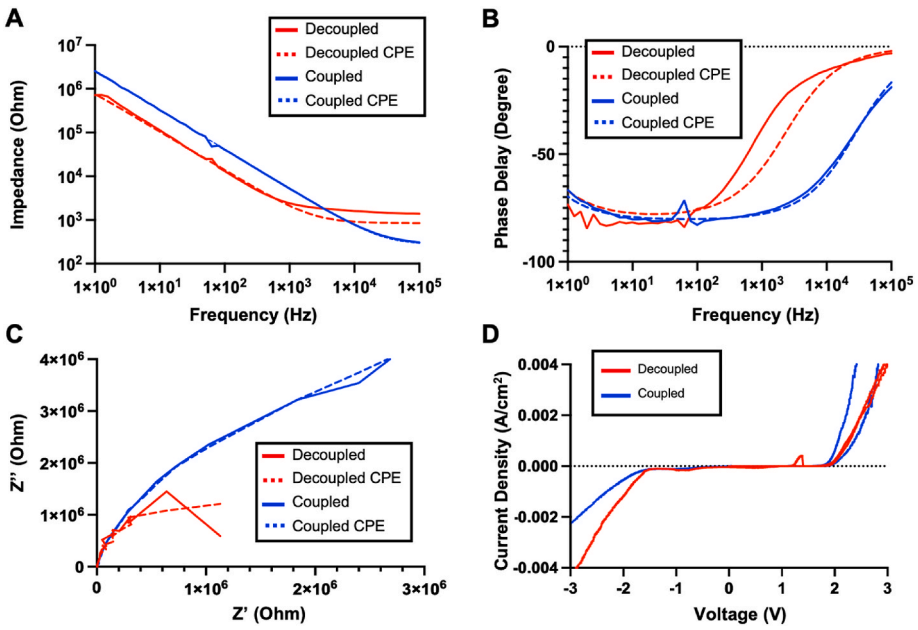


Fig. 3. Electrochemical characterization of electrodes under 'coupled' and 'decoupled' conditions. (A) Impedance magnitude across a range of frequencies, showing lower impedance for the 'de-coupled' condition, indicative of improved charge transfer. (B) Phase delay analysis reveals that the 'de-coupled' condition exhibits phase angles closer to an ideal capacitive response. (C) Nyquist plot displaying a smaller semicircle for the 'decoupled' condition, reflecting reduced charge transfer resistance. (D) Cyclic voltammetry curves demonstrating higher current peaks for the 'decoupled' condition, suggesting enhanced electrochemical activity. Cyclic Voltammetry curves, demonstrating higher current peaks for the 'De-coupled' condition, suggesting enhanced electrochemical activity.

facilitate rapid and efficient ion and electron transport, critical for both cardiotoxicity screening and broader electrochemical applications requiring streamlined charge transfer. The phase behavior (Fig. 3B) further substantiates these findings, with the 'Decoupled' configuration achieving phase angles near −90°, indicative of ideal capacitive behavior. These results underscore the effectiveness of the decoupling treatment in optimizing the electrode-electrolyte interface by promoting uniformity and enhancing its capacitive nature. These enhancements facilitate more effective charge storage and transfer, which are pivotal for precise electrophysiological measurements. Nyquist plots (Fig. 3C) provide complementary evidence, with reduced semicircle diameters observed in the 'Decoupled' state, reflecting a decrease in charge transfer resistance (R_p). Additionally, the near-vertical slope at lower frequencies indicated capacitive-dominated behavior, suggesting that the decoupling treatment minimized surface heterogeneity and roughness while creating a more reactive electroactive surface conducive to efficient charge transfer. Cyclic voltammetry curves (Fig. 3D) showcased increased peak currents for the 'Decoupled' electrodes, signifying enhanced redox activity and improved electrochemical surface accessibility. These findings align with the hypothesis that the decoupling treatment increases the density of electroactive sites and accelerates electron transfer kinetics, thereby improving the overall electrode performance. Table S1 summarizes the quantitative electrochemical parameters, where the 'Decoupled' state exhibited a substantial reduction in polarization resistance (R_p), from 12.77 MΩ to 3.41 MΩ. This improvement was accompanied by an increase in the Constant Phase Element (Y_0), indicative of a larger effective surface area and improved charge storage capabilities. Importantly, the relatively stable CPE exponent (α) across conditions demonstrates that the decoupling process enhances sensitivity and charge transfer efficiency without significantly altering the intrinsic capacitive properties of the interface. These results underscore the superior charge transfer capability of the 'Decoupled' configuration, highlighting its potential to significantly improve the sensitivity and reliability of electrophysiological measurements. These findings establish a strong case for integrating advanced electrochemical topologies into bioelectronic systems, particularly for applications requiring precise and reproducible signal detection, such as cardiotoxicity screening. This study not only validates the efficacy of EIS and cyclic voltammetry as diagnostic tools but also sets a new benchmark for the design of next-generation electrochemical platforms with enhanced performance and sensitivity. Additionally, limit of detection was calculated using the standard signal-to-noise ratio method to further validate the sensitivity improvements in the 'Decoupled' configuration. This analysis revealed a significant enhancement in detection capability, with the limit of detection decreasing from 0.175 MΩ in the coupled condition to 0.040 MΩ in the decoupled condition. This substantial reduction highlights the enhanced sensitivity of the decoupled configuration, enabling more precise detection of subtle changes in polarization resistance.

3.3. *Effects of decoupled electrochemical topology on detection sensitivity in drug-induced cardiotoxicity screening*

Here, we demonstrated the predictive capacity of our nanoMEA-based functional assay in accurately detecting the arrhythmogenic potential of three clinically used medications with known proarrhythmic risks for cardiotoxic and arrhythmic effects in hiPSC-CMs. As described earlier, we enhanced the signal linearity and response by decoupling the reference electrode (−) from Nafion, which introduces electrical barriers at the cell-electrode (+) interface. Electrophysiological alterations following compound treatment were analyzed longitudinally at a two-week time point. Robust data were obtained for the three compounds - Ranolazine, Domperidone, and Sotalol - using the reference decoupling configuration, which outperformed the original scheme where Nafion covered the reference electrode (coupled configuration). Each compound was tested at various dose levels based on clinically relevant concentrations observed in patients and compared to in-plate DMSO controls, as shown in Table S2. Changes in electrophysiological conduction and action potential duration are well-established determinants of cardiac arrhythmias (Arita et al., 1976; Campbell et al., 2012; Mironov et al., 2008). Key parameters, including beating period (BP), field potential duration (FPD), spike slope, and spike amplitude, were analyzed and compared using a double subtraction method. This involved first calculating the pre- and post-drug treatment differences,

followed by subtracting the corresponding changes in the DMSO group. One-way ANOVA with Dunnett's post-hoc analysis showed statistically significant changes in BP for all compounds, FPD for Domperidone, and both spike slope and amplitude for Ranolazine (Fig. S1). Under the de-coupled condition, BP analysis showed a significant increase in BP for all tested compounds (Fig. 4A–C). Ranolazine, Sotalol, and Domperidone are known to prolong the QT interval by binding to the voltage-gated potassium channel hERG, thereby reducing rapid potassium rectifier currents or intracellular calcium levels (Jankelson et al., 2020; Vormfelde et al., 2020). At high doses of all compounds, cardiac rhythm was completely arrested, and no signals were recorded. For FPD values, Domperidone demonstrated a prolongation effect under the de-coupled condition at concentrations 10 to 30 times lower than those required under the coupled condition. Additionally, in the de-coupled condition, a 5 μM treatment of Sotalol significantly prolonged the BP, although no changes in spike slope were detected. By contrast, in the coupled condition, the beat period prolongation was observed only at 20 times the treatment concentration, and spike slope changes were recorded to decrease at 50 μM treatment of Sotalol.The enhanced sensitivity of the de-coupled condition was particularly evident in its ability to detect changes at lower drug concentrations. For example, Ranolazine treatment significantly prolonged the beat period values starting at 5 μM under the de-coupled condition, while no variation was observed in the coupled state below 50 μM. Ranolazine exerts its effects

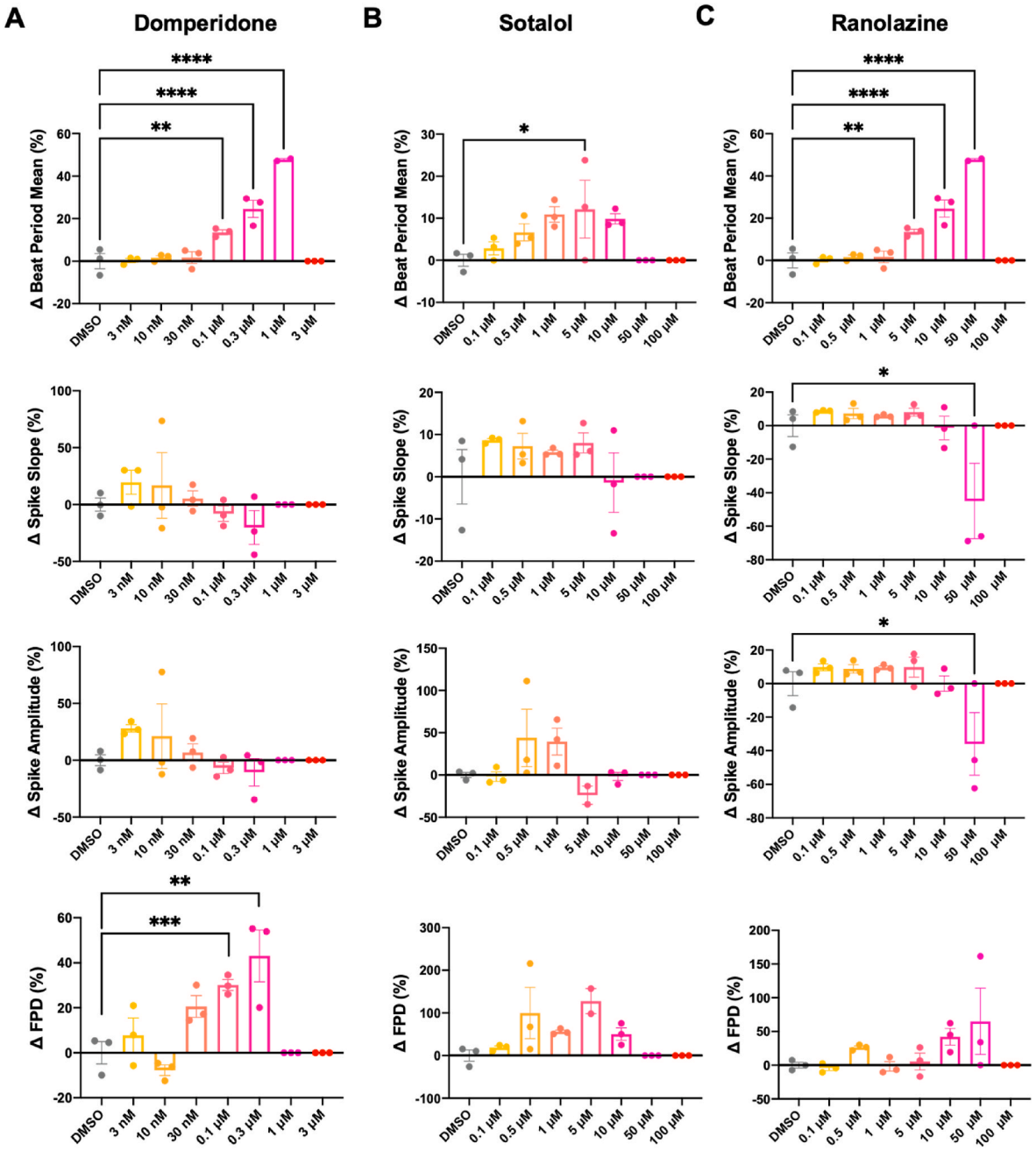
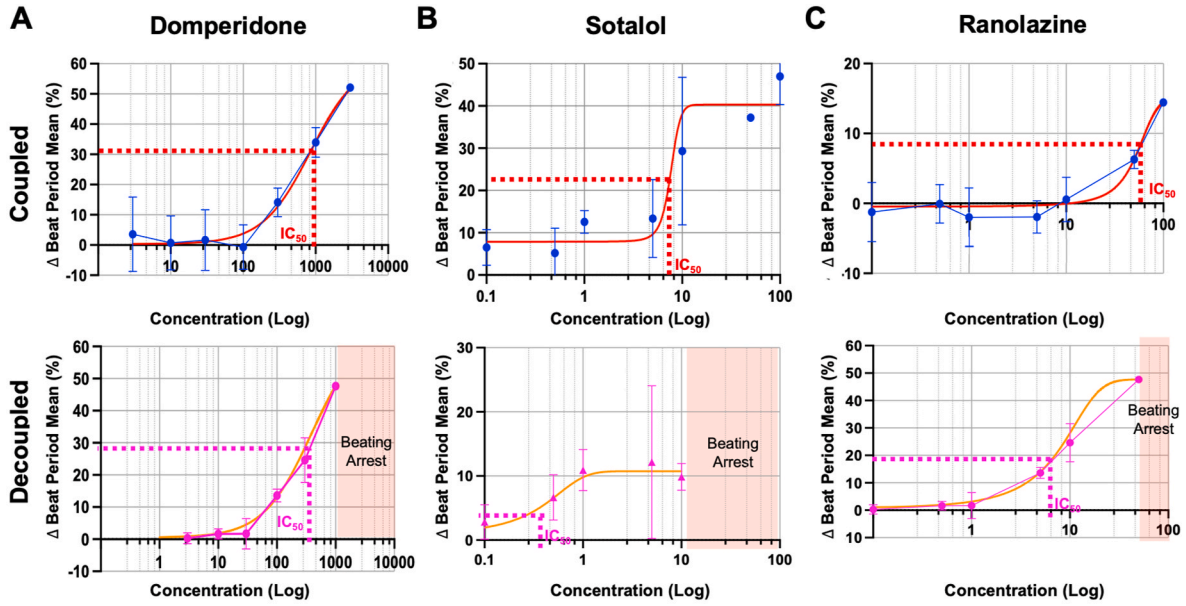


Fig. 4. Evaluation of the pharmacokinetic parameters of cardiac proarrhythmic compounds under decoupled conditions. Bar graphs represent the percentage change in beat period mean, field potential duration (FPD), spike slope, and spike amplitude at various concentrations of three test compounds (A) Domperidone, (B) Sotalol, and (C) Ranolazine relative to the DMSO control. Data points indicate individual measurements, while bars show the mean ± standard deviation. Statistical significance is denoted by asterisks (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$), highlighting the dose-dependent effects of the compounds on cardiac electrophysiology parameters under coupled conditions.

by inhibiting the persistent or late inward sodium current (I_{Na}) as well as the rapid delayed rectifier potassium current (I_{Kr}) in cardiac muscle. This inhibition reduces spike amplitude due to diminished inward sodium currents (Antzelevitch et al., 2011). In this study, both spike slope and amplitude parameters were significantly reduced at a 50 μM treatment concentration of Ranolazine, regardless of whether the conditions were coupled or de-coupled. Following this, we evaluated the pharmacological impact of three cardiac drugs on the rhythmic contractions of hiPSC-CMs under different conditions (Fig. 5). The comparison between coupled and de-coupled conditions revealed distinct differences in drug responsiveness, as reflected in the shift of concentration-response curves and variations in half-maximal inhibitory concentration (IC_{50}) values. These curves provided quantitative measures of drug efficacy and potency, which are essential for understanding the pharmacodynamics of these compounds. Analytical rigor was ensured through the implementation of the ROUT method for outlier detection, applied with an aggressiveness setting of Q = 0.2%. A standard regression model was used to generate sigmoidal dose-response curves, allowing for precise characterization of the pharmacological effects of each drug. The de-coupled condition displayed a marked decrease in IC_{50} values for all tested drugs, indicating increased sensitivity of the cardiac cells to these compounds in the absence of mechanical coupling. For Domperidone (Fig. 5A), IC_{50} values decreased significantly from 846.5 μM under coupled conditions to 379.4 μM in the de-coupled state. Similarly, Sotalol (Fig. 5B) exhibited a similar trend with IC_{50} values shifting from 7.61 μM to 0.27 μM, while Ranolazine's (Fig. 5C) IC_{50} decreased from 53.08 μM to 5.89 μM. These findings highlight the differential drug efficacies observed under varying recording environments and suggest the potential influence of the measurement scheme on the pharmacodynamics of these drugs. The results further validate the de-coupled condition as a more sensitive and reliable method for assessing drug responses in cardiac cells.

3.4. *Effects of decoupled electrochemical topology on local extracellular action potential signals*

FPD is a commonly measured electrophysiological parameter in cardiomyocyte MEA studies (Clements and Thomas, 2014). However, conventional FPD measurements often lack the ability to capture detailed kinetics of action potential waveforms due to the limited number of electrodes per well and the reliance on identifying a "golden electrode" (Gintant et al., 2020). Our LEAP analysis addresses these limitations and reveals that action potential waveform kinetics closely follow the trends observed in BP and FPD changes. This action potential assay allows us to indirectly analyze key action potential waveform parameters such as repolarization times (Action potential duration 30, 50, and 90; APD30, APD50, APD90) and action potential triangulation (Fig. 6 for the 'decoupled' condition and Fig. S2 for the 'coupled' condition). We compared AP waveform parameters before and after treatment with the IC_{50} concentrations calculated in Fig. 5 for both 'coupled' and 'decoupled' conditions. Ranolazine is known to inhibit inward rectifier potassium channels and prolong action potential duration (Antzelevitch et al., 2011). Domperidone prolongs action potential duration by blocking hERG potassium channels in the heart (Claassen and Zunkler, 2005). Consistent with the significantly prolonged QT interval observed in BP and FPD changes, Domperidone at a higher dose



D

Compound	Known Risk	Control	IC_{50} Concentrations (μM)	
			Coupled	Decoupled
Domperidone	Medium	DMSO 0.1%	846.5	379.4
Sotalol	High		7.61	0.27
Ranolazine	Low		53.08	5.89

Fig. 5. Concentration-response relationships for selected cardiac drugs under coupled and decoupled conditions. Percentage change in mean cardiac beat period plotted against drug concentration (log scale) for Domperidone (A), Sotalol (B), and Ranolazine (C). Each drug under both 'coupled' (top panels) and 'decoupled' (bottom panels) conditions, with error bars indicating variability across independent measurements. Horizontal dashed lines highlight the derived IC50 thresholds, while shaded regions indicate concentrations leading to beating arrest. (D) Summary table listing known risk categories for each drug (Domperidone, Sotalol, Ranolazine) and their IC50 values (in μM) under coupled and decoupled conditions, as well as under control conditions (0.1% DMSO).

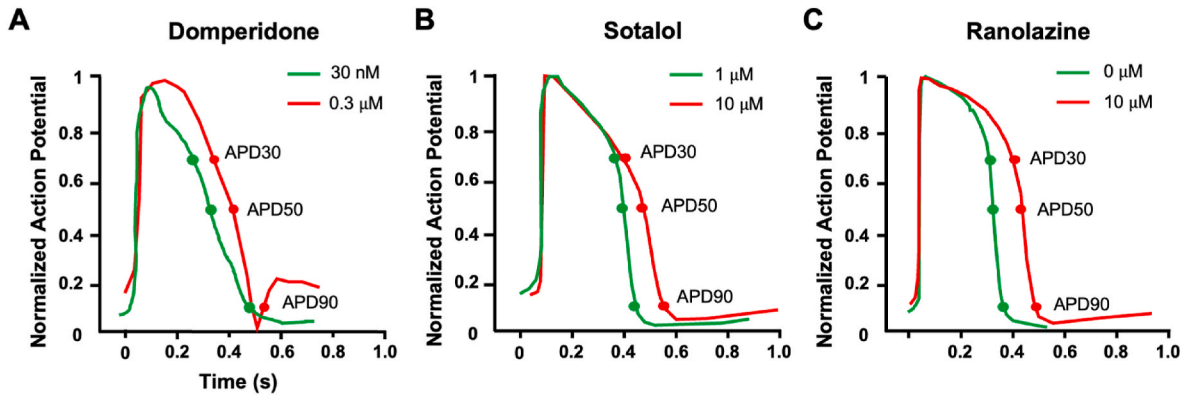


Fig. 6. Analysis of local extracellular action potential signals demonstrating compound-dependent changes in action potential repolarization under decoupled conditions. Amplitude-normalized action potential waveforms, and action potential duration (APD) graphs illustrate the effects of each compound. Action potential signals are shown for low-dose (green) and high-dose (red) conditions to highlight dose-dependent changes for (A) Domperidone, (B) Sotalol, (C) Ranolazine. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(0.3 μM, higher than IC_{50} from Fig. 5 for the 'de-coupled' condition) led to a prolongation of AP duration (Fig. 6A) compared to a lower dose (30 nM, lower or around IC_{50} from Fig. 5 for the 'de-coupled' condition). Sotalol reduces calcium influx, which decreases intracellular calcium levels and subsequently diminishes the electrical signals required for cardiomyocyte contraction. Consistent with this mechanism, a previous MEA study identified Sotalol as a high torsades de pointes (TdP) risk agent (Kussauer et al., 2019). Our AP analysis corroborated these findings, showing that high-dose Sotalol (10 μM, higher than IC_{50} from Fig. 5 for the 'decoupled' condition) slightly increased the repolarization phase (Fig. 6B) compared to low-dose treatment (1 μM, lower or around IC_{50} from Fig. 5 for the 'de-coupled' condition). A previous study reported FPD increases at Ranolazine concentrations of 3–30 μM (Millard et al., 2018). Consistent with this, our AP analysis showed that high-dose Ranolazine (10 μM, higher than IC_{50} from Fig. 5 the 'decoupled' condition) resulted in a more pronounced increase in the repolarization phase (Fig. 6C) compared to the control (DMSO-treated, lower than IC_{50} from Fig. 5 for the 'decoupled' condition). These results demonstrate the effectiveness of our innovative cardiotoxicity screening platform, which utilizes a decoupled Nafion-coated NanoMEA with AP analysis, for assessing the effects of proarrhythmic test drugs on hiPSC-CMs. Our platform successfully reproduced the effects of several previously identified high-risk agents on BP and FPD, while AP analysis provided deeper insights into the drug effects on hiPSC-CM activity.

4. Conclusion

This study provides a comprehensive evaluation of electrode performance and pharmacological responsiveness in a bioelectronic interface using hiPSC-CMs. By comparing 'Coupled' and 'Decoupled' configurations of Nafion-modified electrodes, we demonstrated that the decoupled configuration significantly improves charge transfer dynamics, reduces polarization resistance, and enhances the electroactive surface area. These improvements established a more homogenous and efficient electrochemical environment evidenced by electrochemical impedance spectroscopy and cyclic voltammetry measurements. And it optimized the sensitivity and reliability of the NanoMEA system for cardiotoxicity assessments. Pharmacological assessments further validated the utility of the NanoMEA system for drug screening applications. The decoupled configuration demonstrated heightened sensitivity, as indicated by reduced IC_{50} values and precise detection of drug-induced electrophysiological changes, including alterations in BP, FPD, spike slope, and spike amplitude. These findings highlight the platform's ability to detect subtle proarrhythmic risks and reliably reproduce the known effects of high-risk drugs on cardiomyocyte electrophysiology. This study underscores the critical importance of electrochemical topology in the design of next-generation bioelectronic devices. By integrating nanopatterned Nafion interfaces and decoupled reference electrodes, the NanoMEA system improves electrode-cell coupling, reduces noise, and enhances signal quality. These advancements represent a significant step forward in developing reliable and sensitive platforms for drug screening and cardiac electrophysiology, paving the way for further innovations in preclinical drug development and cardiac research.

CRediT authorship contribution statement

Byunggik Kim: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jong Seob Choi:** Writing – review & editing, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Yiguang Zhu:** Writing – original draft, Investigation, Formal analysis. **Juhyun Kim:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Ye Seul Kim:** Writing – review & editing, Methodology, Investigation. **Andres Parra:** Writing – original draft, Investigation, Formal analysis. **Paul A. Locke:** Writing – review & editing. **Jae Ho Kim:** Writing – review & editing. **Todd Herron:** Writing – review & editing, Validation, Methodology. **Deok-Ho Kim:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

Deok-Ho Kim is a co-founder and scientific advisory board member at Curi Bio, Inc.

Todd. J. Herron is a consultant and scientific advisory board member at StemBioSys, Inc.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bios.2024.117082.

Data availability

Data will be made available on request.

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